

Serving Size Scoops

Unit 2: Fraction Detective Agency · Lesson 2-2 · EduWonderLab

UNIT 2 · LESSON 2-2 · TEACHER NOTES

Standard 6.NS.1

FORMAT Recipe partner game	TIME 25–30 min
GROUPING Pairs (Modeler & Solver)	TOPIC Divide Whole Numbers by Fractions
STANDARD 6.NS.1	PREP LEVEL Light — print recipe cards and fraction strip templates

Learning goal *I can divide a whole number by a fraction to find how many servings fit.*

Teacher setup

- Print one recipe set per pair. Assign roles: Modeler draws, Solver computes.
- For each card, the Modeler sketches the servings; the Solver uses the reciprocal.
- Partners compare the picture and the calculation, then switch roles.

Materials

Recipe cards, Fraction strip templates, Reciprocal reminder card, Pencil.

Differentiation & language support

- Reciprocal reminder card: “Dividing by a fraction = multiplying by its reciprocal.” (SPED)
- Sentence stem: “There are ___ servings because $__ \div __ = __$.”
- ESOL: use food words students know (scoops, servings, cups).

Key vocabulary (English / Español):

Term	Español	What it means
reciprocal	recíproco	a fraction flipped upside down
serving	porción	one helping of food

Quick teacher check: *Have a pair show the reciprocal step for Card 4 ($5 \div 2/3$).*

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Name: _____ Date: _____ Class: _____

Your goal: *I can divide a whole number by a fraction to find how many servings fit.*

What you need

Recipe cards, Fraction strip templates.

How to play

1. Pick a recipe card. The Modeler sketches the whole amount split into servings.
2. The Solver rewrites the division as multiplying by the reciprocal.
3. Both partners check the picture matches the answer.
4. Write how many servings and switch roles.

Recipe Cards

Find how many fractional servings fit inside each whole-number amount. Draw it, then solve with the reciprocal.

1. 4 cups of soup ÷ 1/2-cup servings	2. 6 cups of rice ÷ 1/3-cup servings
3. 3 cups of nuts ÷ 1/4-cup servings	4. 5 cups of trail mix ÷ 2/3-cup servings
5. 8 cups of cereal ÷ 2/5-cup servings	6. 2 cups of sauce ÷ 3/4-cup servings
7. 10 cups of juice ÷ 5/6-cup servings	8. 7 cups of oats ÷ 1/2-cup servings
9. 9 cups of flour ÷ 3/4-cup servings	10. 4 cups of broth ÷ 2/3-cup servings

Record your work (model + reciprocal step):

Explain your thinking

Explain why dividing by a fraction gives more servings than dividing by a whole number.

Sentence frame: *Dividing by ___ gives a ___ answer because the servings are ___.*

★ **Challenge:** Card 4 and Card 6 do not come out even. Explain what the “ $\frac{1}{2}$ ” and “ $\frac{2}{3}$ ” of a serving mean.

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. 4 cups of soup $\div$ $\frac{1}{2}$ -cup servings Answer: $4 \div \frac{1}{2} = 8$ servings	2. 6 cups of rice $\div$ $\frac{1}{3}$ -cup servings Answer: $6 \div \frac{1}{3} = 18$ servings
3. 3 cups of nuts $\div$ $\frac{1}{4}$ -cup servings Answer: $3 \div \frac{1}{4} = 12$ servings	4. 5 cups of trail mix $\div$ $\frac{2}{3}$ -cup servings Answer: $5 \div \frac{2}{3} = 7 \frac{1}{2}$ servings
5. 8 cups of cereal $\div$ $\frac{2}{5}$ -cup servings Answer: $8 \div \frac{2}{5} = 20$ servings	6. 2 cups of sauce $\div$ $\frac{3}{4}$ -cup servings Answer: $2 \div \frac{3}{4} = 2 \frac{2}{3}$ servings
7. 10 cups of juice $\div$ $\frac{5}{6}$ -cup servings Answer: $10 \div \frac{5}{6} = 12$ servings	8. 7 cups of oats $\div$ $\frac{1}{2}$ -cup servings Answer: $7 \div \frac{1}{2} = 14$ servings
9. 9 cups of flour $\div$ $\frac{3}{4}$ -cup servings Answer: $9 \div \frac{3}{4} = 12$ servings	10. 4 cups of broth $\div$ $\frac{2}{3}$ -cup servings Answer: $4 \div \frac{2}{3} = 6$ servings

Teaching notes & look-fors

- Cards 4 and 6 have mixed-number answers: $7 \frac{1}{2}$ and $2 \frac{2}{3}$ servings.
- Reciprocal step: $a \div \frac{b}{c} = a \times \frac{c}{b}$.